

ls ---> used to list files and dirs

ls -l --> list files and dirs with long format

touch test1 ---> used to create empty file

ls -lt ---> recently created files will be displayed at beginning

ls -lrt --> recently created files will be displayed at the end

pwd ---> present working directory

mkdir dirname ----> create dir

cd dirname ----> change directory

cd dir/dir2/

cd .. --> used to come out of directory

cd ../../.. --->

mkdir -p temp1/temp2/temp3 ---> create complete directory structure

-
cat filename ---> used to check content of a file

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vi is used to edit file

vi filename

esc --> i ---> insert

esc --> :wq! ---> save content of a file (write and quit)

esc ---> :q! --> quit without saving content

how do replace aprticular string or word in a file?

esc :%s/current-string/new-string/g

esc :%s/linux/windows/g

Need to replace only in 4th line

esc:4s/linux/windows/g

need to replace from 2nd to 4th line

esc:2,4s/linux/windows/g

how do you replace from 2nd line to end of a file

esc:2,\$s/linux/windows/g

how do set line numbers

esc : set nu --> set line numbers

esc : set nonu ---> remove line number

need to move cursor to particular line

esc :line_number

esc : 4 ---> cursor will move to 4th line

delete particular line in vi

esc press dd

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cp is used to copy file

cp file1 file2 ----> file1 will be copied to file2

cp test1 temp1/temp2/temp3/

copy dir to another dir

cp -r dir1 dir2

cp test1 file1 test3 temp ---> copy multiple files or dirs to another dir

how do rename file or dir?

mv file1 file2

mv file1 dir2

mv file1 file2 file3 dir1 ---> move multiple files to directory

install tree command
sudo yum install tree

rwxr-xr--
7 5 4

chmod is used to change permission of a file or dir

chmod 777 filename

chmod -R 777 dir

all subdirectories and files permission will be changed to 777

u--> user, g--> group and o--> others

chmod u+rwx filename

chown --> used change ownership of a file or a dir

chmod newowner filename

if new owner belongs to different group

chmod newowner:groupname filename

df -h . ----> used to check disk size

. indicates present disk

du -sh filename (disk usage)

used to check size of a file

du -sh * ---> all files and dirs size in present directory

echo "hi how r u" is used to print

echo -e "hi\nhow r u"

> (redirect)

used to store output of a command to file

ls -lrt > log

>> (append)

used to attach output of a command to end of a file

echo "this is linux class" >> log

=====

wc filename

lines words chars filename

wc -l log ---> number of lines

wc -w log ---> words

wc -c log --> chars

"head" is used to display 1st portion of a file

head -3 filename ---> 1st 3 lines

head -12 filename ---> 1st 12 lines of a file

head filename ---> 10 lines by default

tail is used to display last portion of a file

tail -3 filename --> last 3 lines

tail -5 filename ----> last 5 lines

tail filename -----> last 10 lines by default

| (pipe) is used to give output of one command as input to next command

ls -lrt | wc -l

head -4 log | tail -1 --> display 4th line

how do you display 99th line of a file

head -99 file | tail -1

display 4th to 7th line

head -7 file | tail -3

count number of words in 99th line of a file

head -99 log | tail -1 | wc -w

display 2nd last line of a file

tail -2 log | head -1

grep --> pattern search

used to search string in file

grep "pattern" filename

grep -w "string" filename ----> search only specified word

grep -i "Linux" test_1 ---> i --> case insensitive

how do you search multiple pattern?

grep -e "linux" -e "windows" test_1

or
egrep

how do you list filenames if files have particular word or string

grep -R -l "linux" *

how do you print lines which don't have specified pattern

grep -v "linux" test_1

list lines if starts with particular pattern

grep "^pattern" filename

grep "s\$" test_1 ---> list lines ends with particular pattern

get command to print lines which start with "M" and ends with "s"

grep -c "linux" test_1 ---> count number lines which have specified pattern

how do you replace string in a file?

sed --> used to replace string in a file
sed 's/linux/unix/g' filename

```
sed -i 's/linux/unix/g' test_1
sed '4s/linux/unix/g' ---> replace string in 4th line only
sed '2,6s/linux/unix/g' ---> from 2nd to 6th line
sed '$s/linux/unix/g' ---> replace string only in the last line
delete particular line
sed '4d' test_1 --> delete 4th line of a file
sed '$d' test_1 --> delete last line
sed '2,6 d' test_1 --> delete 2nd to 6th line
sed '1d;3d' test_1
how do you print particular line?
sed -n '3p' test_1 ---> print 3rd line
sed -n '$p' test_1 ---> print last line
sed -n '2,6p' test_1 ---> print 2nd to 6th line
sed -n '2p;6p' test_1 ---> print 2nd to 6th line
```

```
cut --> used to cut file colounwise
cut -d " " -f1 data --> 1st coloumn
cut -d " " f1,3 data ---> 1st and 3rd coloumn
cut -d " " f3-6 data ---> 3rd to 6th column
```

```
awk is used to cut file rowwise and coloumn wise but we use awk command to cut
file coloumn wise
```

```
awk -F " " '{print $1}' data ----> 1st coloumn
```

```
$1 ---> 1st coloumn
```

```
$2 --> 2nd coloumn
```

```
$NF ---> last coloumn
```

```
$(NF-1) --> 2nd last coloumn
```

```
awk -F " " '{print $NF}' data ---> last coloumn
```

```
awk -F " " '{print $(NF-1)}' data --> 2nd last coloumn
```

```
find ---> used to find location of a file
```

```
find . -iname "filename"
```

```
i --> case insensitive
```

list files which are modified 3months back

```
find . -type f -mtime +90
```

```
find . -type d -mtime +90
```

```
find . -mtime +90 -->both files and dirs
```

list files which are modified within 3months

```
find . -type d -mtime -90
```

find files which are modified with in 50mins

```
find . -type f -mmin -50
```

find files which modified 50mins ago

```
find . -type f -mmin +50
```

find files which have 777 permission

```
find . -type f -perm 777
```

find all empty files

```
find . -type f -empty
```

-----command to list all non empty files-----

```
find . -maxdepth 1 -iname "test1"
```

maxdepth is used to restrict automatic recursive

maxdepth 1 ---> 1st level dir

maxdepth 2 ---> 2nd level dir

delete all files which are modified 3 months ago

```
find . -type f -mtime +90 | xargs rm -rf
```

xargs is used to pass output of one command to next command as arguments

softlink

it's shortcut to a file. if you make any changes in the actual file, it will get reflected in the softlink

if you delete actual file, softlink will not work

```
ln -s /home/ec2-user/file1 link1
```

hardlink

```
ln /home/ec2-user/file1 hard1
```

--

diff b/w hard link and softlink

if I delete actual file, softlink will not work

if I delete actual file, hardlink works fine because it points to inode of a file (inode is uniq number indicates memory address)

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Assignment:

1. command to list files if their size is more than 1MB

2. use -exec option in find command to delete files

ps --> used list current running processes on the system

need to check whether particular process is running or not

ps -ef | grep "processname"

3. need to list all processes started by particular user

stop process (forceful stop)
kill -9 PID

4. kill all process started by particular user

5. kill all process by name

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how do stop process gracefully

sudo service name stop
sudo service name start
sudo service name restart

6. get systemctl command

.bashrc

what is .bashrc or .bash_profile?

an auto-execute file which gets executed automatically as soon as log into system

what is diff b/w .bashrc , bash_profile and bash_history?

umask is used to set default permission on system.
it should run with root user

umask 000 ---> whoever creates file or dir will have full permission

umask 777 ---> no permission

rw-r-xr-x = 755

777
755

022

umask 0222

r-xr-xr-- = 554

777
554

223

umask 233

su (switch user or superuser)
is used to login as other user or root user

su - username
password:

sudo (superuser does)

sudo ---> used to execute command with root permission

sudo yum install vim wget

add user ABC abd and give sudo permission

sudo useradd ABC
sudo passwd ABC
password:
password:

sudo su - (login as root user from ec2-user)

vi /etc/sudoers.d/90-cloud-init-users

go to last line of this file

give same permission for ABC also like ec2-user
you need to add another line and change ec2-user to ABC
save file
ABC also will have sudo permission

more command is used to display content of a file page wise

more filename

less ---> used to display content of a file pagewise, but you can scroll up and scroll down using up and down arrows

less filename

(press q to quit)

date --> used to print system date and time
date +%m

uniq --> print onlu uniq data and removes deplicates. use this along with sort
sort --> used to sort data --> ascending sort
sort -r filename ---> decending sort

cat test | sort | uniq

we need to use uniq command always with sort command

how do you check linux version

uname -v ---> only version

uname -a ---> all info

who ---> who all logged into system
whoami ---> check ur username
hostname --> hostname or ip of current server

ssh --> used to login to remote server and uses port 22

ssh username@server
password:

ssh -i aws-keys/devops-june-21.pem ec2-user@172.31.21.0

you need to private IP

run command on remote server

ssh -i aws-keys/devops-june-21.pem ec2-user@172.31.21.0 "ls"

1. install winscp
2. using winscp copy pem file to server1
3. login to server1 using putty
4. change permission to 400 using chmod 400 filename.pem
5. ssh -i filename.pem ec2-user@server2-private-ip

aws ec2-instance

copy file

scp -i aws-keys/devops-june-21.pem data ec2-user@172.31.21.0:/home/ec2-user/

copy dir

scp -i aws-keys/devops-june-21.pem -r temp1/* ec2-user@172.31.21.0:/home/ec2-user/temp1/

```
scp -i aws-keys/devops-june-21.pem  
ec2-user@172.31.21.0:/home/ec2-user/temp1/file1 .
```

rsync ---> used to copy files or dirs within server and also between servers

```
rsync -i key.pem dir1 username@server2:/home/ec2-user/dir1
```

diff rsync and scp

while copying while data from one server to another, if copy is stopped in between due to network issue. if use scp command, it will start copying from beginning
if use rsync command, it will start copying from where it was stopped (because it will compare and copy and it's slower than scp)

```
-----  
-----  
-  
  
-----  
-----
```

ping is used to check server is running or not. ping uses ICMP protocol

(you can server is shutdown or not)

ping ipaddress or servername

```
-----  
--
```

telnet --> used to break firewall. it uses port 23. used to authenticate account credentials on remote server

telnet IPAddress or servername

username:

password:

```
-----  
-----
```

how do you setup passwordless connection between servers ?

we need to run "ssh-keygen -t rsa" on server1 and generate rsa key.
take this public key and copy it to another server to file called
authorized_keys under .ssh directory. save it

next time when you login from server1 to server2, it will be passwordless connection

steps:

login to server1

```
ssh-keygen -t rsa
```

(keep pressing enter key till key get generated)

```
cat /home/ec2-user/.ssh/id_rsa.pub
```

(copy this key)

login to server2

```
cd .ssh
```

```
vi authorized_keys
```

```
paste key at the end of a file and save it
```

```
go to server1 and run "ssh ec2-user@server2" (check logged in to server2  
without password)
```

```
-----  
-----
```